

LOW ACRYLONITRILE NBR PATENT LANDSCAPE REPORT

1. ABSTRACT

This report reviews patents relevant to low-acrylonitrile nitrile butadiene rubber (NBR) synthesis and use, with emphasis on compositions and processing routes that preserve rubber-grade performance while reducing acrylonitrile content relative to conventional higher-ACN grades. The patent set does not disclose a dedicated low-ACN NBR polymerization process; instead, it shows how NBR is selected, blended, or processed in end-use formulations where acrylonitrile content is controlled as a material property. EP3828231A1 discloses a halogen-free cable sheath composition based on an EVA/NBR/AEM blend, explicitly allowing NBR grades with acrylonitrile contents of 31-35 wt%, 36-41 wt%, or 42-46 wt%, thereby covering a lower-ACN rubber window within a performance-driven cable formulation. US10385647B2 focuses on swellable packer elastomers and uses a nitrile elastomer together with butadiene rubber, sulfur cure, silica, and powder super absorbent polymer; the examples emphasize high-temperature/high-pressure brine swelling, and the disclosure notes a hydrogenated nitrile elastomer with 70-91% hydrogenation and Mooney viscosity ML(1+4) of 47-72, but it is not a low-ACN synthesis patent. US10731020B2 is a vulcanization-molding rubber composition patent centered on hollow particles and base rubber selection; it lists nitrile rubber among many possible base rubbers and specifies a base-rubber Mooney viscosity range of 5-90 at 100°C, again without a dedicated NBR polymerization route. Across the set, the dominant trends are formulation-based performance tuning, use of polar elastomer blends, and processability control via viscosity, filler loading, and cure conditions rather than monomer-feed synthesis. For low-ACN NBR development, the evidence supports selecting NBR grades in the 31-41 wt% acrylonitrile range when balancing oil resistance, flexibility, and cold performance, then optimizing downstream compounding and cure to maintain mechanical integrity.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP3828231A1
- Patent Title: ... composition comprising nitrile rubber and ethylene methyl acrylate copolymer
- Assignee: Not disclosed
- Jurisdiction: EP

- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose NBR polymerization; it discloses a crosslinked cable sheath composition formed from a polymer blend of EVA, NBR, and AEM with a crosslinking agent and flame-retardant filler. The NBR is selected as a commercial rubber grade within specified acrylonitrile-content windows for performance tuning in a halogen-free cable formulation. - Vinyl acetate content: less than 40 wt% — Claims - Acrylonitrile content: 42 to 46 wt% — Claims - Acrylonitrile content: 36 to 41 wt% — Claims - Acrylonitrile content: 31 to 35 wt% — Claims - Polymer blend composition: 15 to 30 parts by weight EVA, 35 to 45 parts by weight NBR, 25 to 50 parts by weight AEM per 100 parts by weight polymer blend — Detailed Description - Crosslinking agent: 0.5 to 10 parts by weight — Detailed Description - Crosslinking coagent: 0.5 to 7 parts by weight — Detailed Description - Plasticizer: 5 to 20 parts by weight — Detailed Description - Processing agent: 0.5 to 5 parts by weight — Detailed Description - Antioxidant: 0.5 to 7 parts by weight — Detailed Description - Antiozonant: 0.1 to 5 parts by weight — Detailed Description - Inorganic additive: 1 to 20 parts by weight — Detailed Description - Flame-retardant filler: 120 to 180 parts by weight — Detailed Description

Relevant Experimental Evidence - Example 1: No structured example data was extracted; the disclosure is primarily compositional and claim-based. - Example 2: No structured example data was extracted; the patent emphasizes formulation ranges rather than synthesis examples.

Technical Relevance This patent is directly relevant because it explicitly identifies NBR grades with 31-35 wt% and 36-41 wt% acrylonitrile, which fall in the low-to-moderate ACN region of interest. It also shows how low-ACN NBR can be integrated into halogen-free, flame-retardant cable compounds with controlled blend ratios and crosslinking packages.

Patent 2

- Patent Number: US10385647B2
- Patent Title: Swellable rubber compositions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses swellable rubber compositions for downhole packers rather than NBR synthesis. The elastomer system includes a nitrile elastomer and butadiene rubber, cured with sulfur and silica, and loaded with powder super absorbent polymer to achieve controlled swelling in brine under high temperature and pressure. -

Powder super absorbent polymer: about 125 Phr to about 175 Phr — Claims - Powder super absorbent polymer: about 170 Phr to about 210 Phr — Claims - Powder super absorbent polymer: at least about 150 Phr, about 175 Phr, about 200 Phr or about 225 Phr — Claims - Powder super absorbent polymer fraction: at least about 50%, about 60%, about 70%, about 80%, about 90%, or about 95% by weight of total SAP — Claims - Mooney viscosity ML(1+4) (100°C): about 47 to about 72 — Detailed Description - Percent hydrogenation: about 70 percent to about 91 percent — Detailed Description - Percent unsaturation: about 9 percent to about 30 percent — Detailed Description - Swelling: at least about 50%, 60%, 65%, 70%, 75%, 80%, or 90% by volume — Detailed Description - Temperature: about 200 degrees Fahrenheit — Detailed Description - Temperature: about 275 degrees Fahrenheit — Detailed Description - Pressure: at least about 3000, 4000, 5000 or 6000 psi — Detailed Description - Example 1 cure temperature: 330 degrees Fahrenheit — Example 1 - Example 1 cure pressure: 70 psi — Example 1 - Example 1 cure time: 2.5 hours — Example 1 - Example 1 brine temperature: 275 degrees Fahrenheit — Example 1 - Example 1 brine salinity: 16% — Example 1 - Example 1 differential pressure: at least 5000 psi — Example 1 - Example 2 SAP loading: 110 phr, 130 phr, 150 phr, 170 phr — Example 2 - Example 2 test temperatures: 200 degrees Fahrenheit and 275 degrees Fahrenheit — Example 2 - Example 2 brine concentrations: 5% and 16% by volume — Example 2 - Example 3 predicted SAP loading for 75% swell: about 205 Phr (powder SAP), about 190 Phr (medium granular SAP), about 160 Phr (large granular SAP) — Example 3 - Example 3 maximum load: about 170 Phr (medium granular SAP), about 150 Phr (large granular SAP) — Example 3 - Example 4 composition: 205 Phr powder SAP and 10% NBR rubber (Nipol DN3350) — Example 4 - Example 4 composition: 205 Phr powder SAP and 10% HNBR (Zetpol 2030L) — Example 4 - Example 4 swell data: 75.8%, 77.8%, 46.5%, 47.2%, 98.4%, 96.0%, 65.4%, 66.9% — Example 4

Relevant Experimental Evidence - Example 1: A packer using Composition B was cured at 330°F and 70 psi for 2.5 hours and then held in 16% brine at 275°F and about 300 psi, demonstrating high-pressure brine service. - Example 2: Powder, medium granular, and large granular SAPs were compared at 110-170 phr, showing how SAP particle size and loading affect swelling, leaching, and physical properties. - Example 3: The data were used to predict 75% swell loadings, showing powder SAP could reach about 205 phr while granular SAPs were limited by leaching. - Example 4: A 205 phr powder-SAP composition with 10% NBR rubber was compared with 10% HNBR, demonstrating acceptable green strength and strong swell performance. - Example 1A: The method claims specify sealing under about 200°F or 275°F heat, with brine exposure and pressure resistance up to at least about 5000 psi.

Technical Relevance This patent is relevant because it uses a nitrile elastomer in a demanding swellable rubber formulation, showing how nitrile-based rubbers can be compounded for brine resistance, thermal stability, and pressure retention. Although it is not a low-ACN synthesis disclosure, it provides practical formulation and cure conditions that inform downstream use of NBR grades with controlled acrylonitrile content.

Patent 3

- Patent Number: US10731020B2

- Patent Title: Rubber composition for vulcanization molding, process for manufacturing the ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a rubber composition for vulcanization molding based on a base rubber and hollow particles, aimed at achieving lightweight molded products with good dimensional stability and appearance. NBR is listed as one possible base rubber among many elastomers, and the process focuses on kneading, molding, and vulcanization rather than NBR polymerization. - Further expansion ratio: 20 to 80% — Claims - True specific gravity d_1 : 0.02 to 0.40 — Claims - Base rubber Mooney viscosity ML(1+4) (100°C): 5 to 90 — Claims - Hollow particles amount: 0.1 to 30 wt% — Detailed Description - Base rubber amount: 10 to 95 wt% — Detailed Description - Escape-initiation temperature of blowing agent: 110 to 350°C — Detailed Description - Mean particle size of hollow particles: 15 to 100 μm — Detailed Description - Example 1 base rubber Mooney viscosity ML(1+4) (100°C): 54 — Example 1 - Example 1 hollow particle further expansion ratio: 50.0% — Example 1 - Example 1 kneading temperature: 60°C — Example 1 - Example 1 kneading time: about 10 minutes — Example 1 - Example 1 vulcanization molding temperature: 140°C — Example 1 - Example 3 vulcanization temperature: 150°C — Example 3 - Example 3 vulcanization time: 15 minutes — Example 3 - Example 6 injection cylinder temperature: 180°C — Example 6 - Example 6 die temperature: 60°C — Example 6 - Example 6 injection pressure: 15 MPa — Example 6 - Example 11 hollow particle mean particle size (D50): 129.8 μm — Example 11

Relevant Experimental Evidence - Example 1: A composition using a base rubber with Mooney viscosity 54 and hollow particles with 50.0% further expansion ratio gave excellent lightweight effect and appearance. - Example 3: Extrusion followed by vulcanization at 150°C for 15 minutes produced a molded product with good dimensional stability and appearance. - Example 6: Injection molding at 180°C cylinder temperature, 60°C die temperature, and 15 MPa produced a lightweight product with good dimensional stability. - Example 11: Hollow particles with D50 of 129.8 μm still gave an acceptable product, though with a few surface unevennesses. - Comparative Example 1: High Mooney viscosity base rubber caused hollow-particle breakage and loss of lightweight effect.

Technical Relevance This patent is indirectly relevant because it includes nitrile rubber among acceptable base rubbers and highlights how Mooney viscosity, particle expansion ratio, and molding conditions affect processability. It does not target low-ACN NBR synthesis, but it provides useful compounding and vulcanization constraints for NBR-containing rubber products.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP3828231A1 is the only patent that explicitly defines NBR by acrylonitrile-content windows, including 31-35 wt% and 36-41 wt% ACN, which are the closest disclosures to a low-ACN NBR target.
- US10385647B2 and US10731020B2 use nitrile rubber as a formulation component but do not disclose a dedicated acrylonitrile-content range for NBR synthesis; their focus is on end-use performance and processing.
- EP3828231A1 is a halogen-free cable composition with EVA/NBR/AEM blending and high flame-retardant filler loading, whereas US10385647B2 is a swellable packer elastomer with powder SAP and US10731020B2 is a hollow-particle lightweight molding composition.
- US10385647B2 discloses the most detailed cure/service conditions: 330°F cure at 70 psi for 2.5 hours, then brine exposure at 275°F and about 300 psi, with performance targets of at least 5000 psi differential pressure.
- US10731020B2 emphasizes base-rubber Mooney viscosity control (5-90 at 100°C) and hollow-particle further expansion ratio (20-80%), showing a processing-driven rather than composition-driven approach to rubber performance.
- Across the three patents, the recurring formulation trend is blending NBR with other polar or functional components to balance oil resistance, flexibility, swelling, or dimensional stability rather than altering NBR monomer feed during polymerization.

4. CONCLUSION

The evidence indicates that low-acrylonitrile NBR is most clearly represented here as a selectable commercial grade rather than as a newly synthesized polymer. The strongest direct guidance comes from EP3828231A1, which explicitly includes NBR grades at 31-35 wt% and 36-41 wt% acrylonitrile, supporting low-ACN selection for halogen-free cable compounds. US10385647B2 and US10731020B2 reinforce the importance of downstream compounding: nitrile elastomers perform well when paired with appropriate fillers, cure systems, SAP or hollow-particle additives, and tightly controlled viscosity and thermal processing. Based on the disclosed evidence, a practical low-ACN NBR strategy is to select NBR in the 31-41 wt% acrylonitrile range, then tune blend ratios, filler loading, and cure conditions to preserve oil resistance, flexibility, and processability in the final rubber article.

5. REFERENCES

- EP3828231A1 | composition comprising nitrile rubber and ethylene methyl acrylate copolymer | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3828231A1/en>

- US10385647B2 | Swellable rubber compositions | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10385647B2/en>
- US10731020B2 | Rubber composition for vulcanization molding, process for manufacturing the ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10731020B2/en>